

DASHBOARDS REPORT

Customer Churn Prediction — 5aliha 3la Allah Team

Streamlit App & Power BI Dashboard Suite

Dataset	999,999 rows 37 features
Model	LightGBM (AUC-ROC $\approx$ 0.685)
Deliverables covered	Streamlit App (Overview, EDA, Prediction, Retention Plan) + Power BI Dashboard (6 pages)
Prepared by	BI Developer / Deployment Engineer — Team Falcon

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1. Introduction

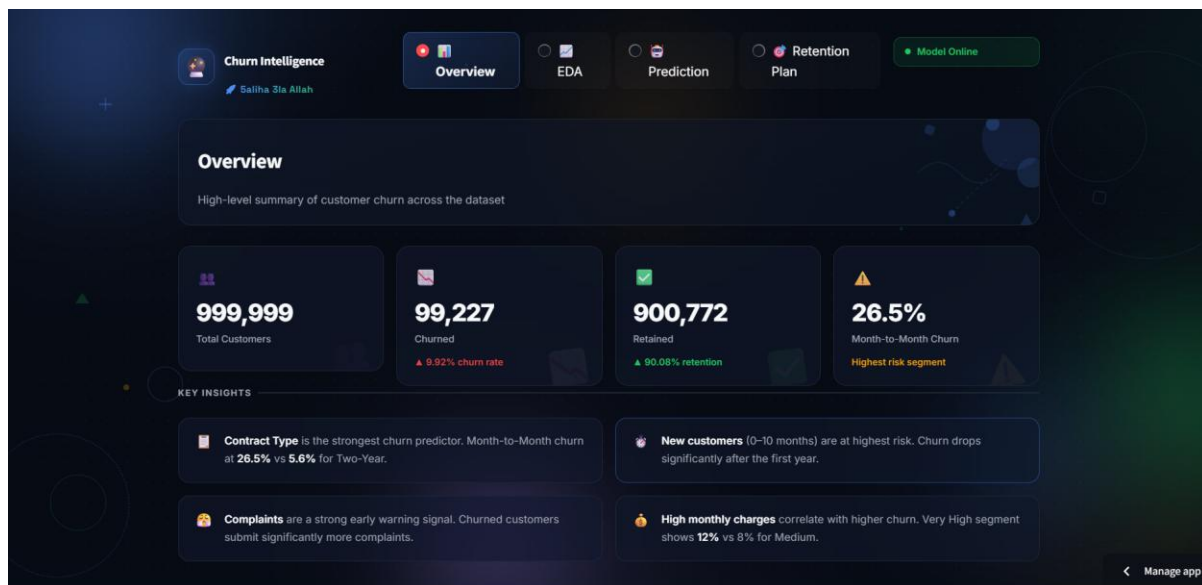
This report brings together every visual deliverable produced for the Customer Churn Prediction project: the four-page interactive Streamlit application (Overview, EDA, Prediction, and the newly added Retention Plan page) and the six-page Power BI dashboard built for deeper, self-service analysis by BI stakeholders.

Both tools share the same underlying dataset (999,999 customer records) and the same LightGBM churn model, but serve different purposes: the Streamlit app is a lightweight, developer-facing prediction and storytelling tool, while the Power BI dashboard is a richer analytical workspace with drill-downs, decomposition trees, and revenue-impact views intended for business reporting.

This document supersedes the earlier single-page Streamlit report — the same Streamlit content is preserved below, now placed alongside the Power BI dashboard work and the new Retention Plan page.

2. Streamlit Dashboard — Overview Page

The app loads the cleaned dataset and the saved LightGBM model once (cached with `@st.cache_data / @st.cache_resource`) and reuses them across all pages. The Overview page surfaces the four headline metrics — total customers, churned count, retained count, and the highest-risk segment — alongside key insight cards extracted during EDA, so a non-technical viewer can grasp the churn story in seconds.



Overview page — headline metrics (999,999 customers, 9.92% churn rate, 26.5% Month-to-Month churn) and key insight cards.

- Contract Type is the strongest churn predictor: 26.5% Month-to-Month churn vs 5.6% for Two-Year contracts.
- New customers (0–10 months tenure) are at highest risk; churn drops significantly after the first year.
- Complaints are a strong early warning signal — churned customers submit significantly more complaints.
- High monthly charges correlate with higher churn (12% for the Very High segment vs 8% for Medium).

3. Streamlit Dashboard — EDA Page

The EDA page reproduces the core findings from the exploratory analysis phase directly inside the dashboard, so stakeholders do not need to open a separate report to see the churn drivers. It plots the raw churn distribution alongside churn rate broken down by contract type.



Churn Distribution (No Churn vs Churn) and Churn Rate by Contract Type — 26.3% Month-to-Month, 12.4% One-Year, 5.6% Two-Year.

Churn Distribution confirms the roughly 9.9% class imbalance in the dataset. Churn Rate by Contract Type shows Month-to-Month customers churn at more than double the One-Year rate and over four times the Two-Year rate — consistent with the correlation analysis performed earlier in the project.

4. Streamlit Dashboard — Prediction Page

The Prediction page lets a user enter a customer's profile through sliders and a contract-type dropdown, then runs that profile through the trained LightGBM model to return a live churn probability with a risk label and contributing risk factors.

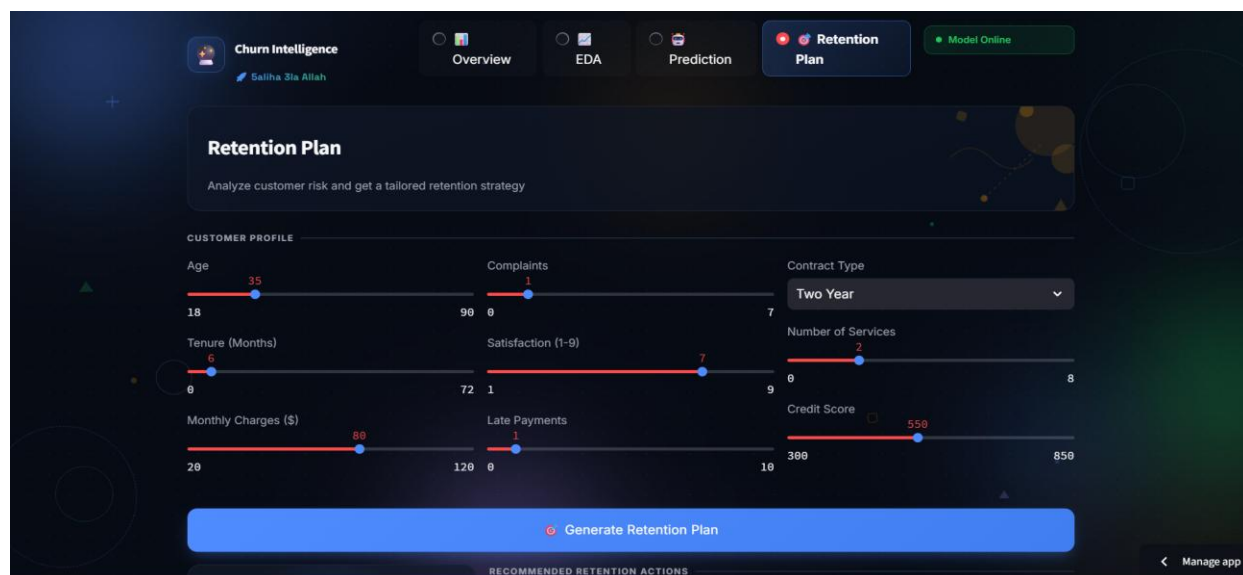


Live prediction example: 35-year-old customer, One-Year contract, 12 months tenure, \$65 monthly charges → 36.7% churn probability, Low Risk.

Since the saved model was trained on scaled data, the sliders use raw, human-readable ranges (Age 18–90, Monthly Charges \$20–120, Credit Score 300–850, etc.) and the input row is built directly into the model's expected feature format before calling `predict_proba`, keeping the interface intuitive while staying faithful to the trained model.

5. Streamlit Dashboard — Retention Plan Page

A new fourth page, Retention Plan, was added to the Streamlit app. It takes the same customer-profile inputs as the Prediction page — now extended with Contract Type, Number of Services, and Credit Score in a dedicated third column — and, instead of only returning a probability, generates a tailored set of recommended retention actions ranked by urgency.



Retention Plan — Customer Profile inputs (Age, Complaints, Tenure, Satisfaction, Monthly Charges, Late Payments, Contract Type, Number of Services, Credit Score) with a 'Generate Retention Plan' action.



Retention Plan output — churn probability gauge, per-factor risk scores, and prioritized recommended actions.

Running the plan for the example profile above (6 months tenure, Two-Year contract, 2 services, credit score 550) returns a 34.6% churn probability labeled Medium Risk, based on the LightGBM model. The page then breaks the prediction down into individual risk factor scores — Complaints (14%), Satisfaction (22%), Tenure Risk (91%), Late Payments (10%), and Contract Risk (10%) — so the highest-contributing factor is immediately visible.

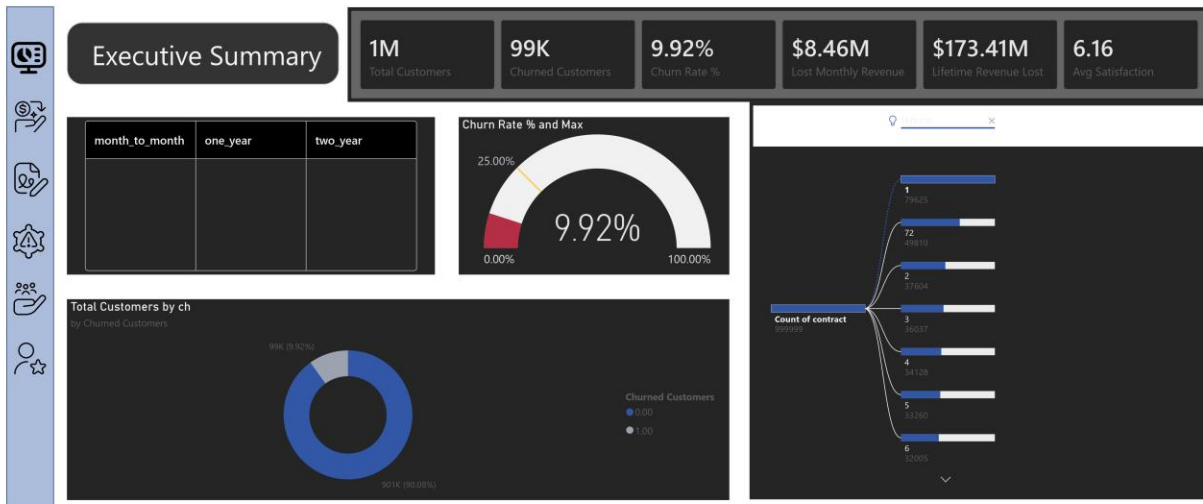
Below the risk breakdown, the page lists concrete, prioritized retention actions, each tagged Urgent, Soon, or Normal:

- **New Customer Onboarding (URGENT)** — customer is only 6 months in, the highest churn-risk window. Recommends a 90-day onboarding program with weekly check-ins and a welcome bonus.
- **Service Bundle Upsell (NORMAL)** — customer only has 2 services. Recommends a bundle deal with 3+ services at a 15% discount, since more services correlates with lower churn risk.

A summary strip totals the recommended actions by priority (e.g. 1 Urgent, 0 Soon, 1 Normal) and nudges the user to start with urgent items first — turning the model's raw probability into an actionable retention playbook for account teams.

6. Power BI Dashboard — Executive Summary

The Power BI workbook opens on an Executive Summary page aimed at leadership: a single row of KPI cards followed by a contract-type breakdown table, a churn-rate gauge, a churn donut, and a decomposition tree for drilling into the drivers of churn count.

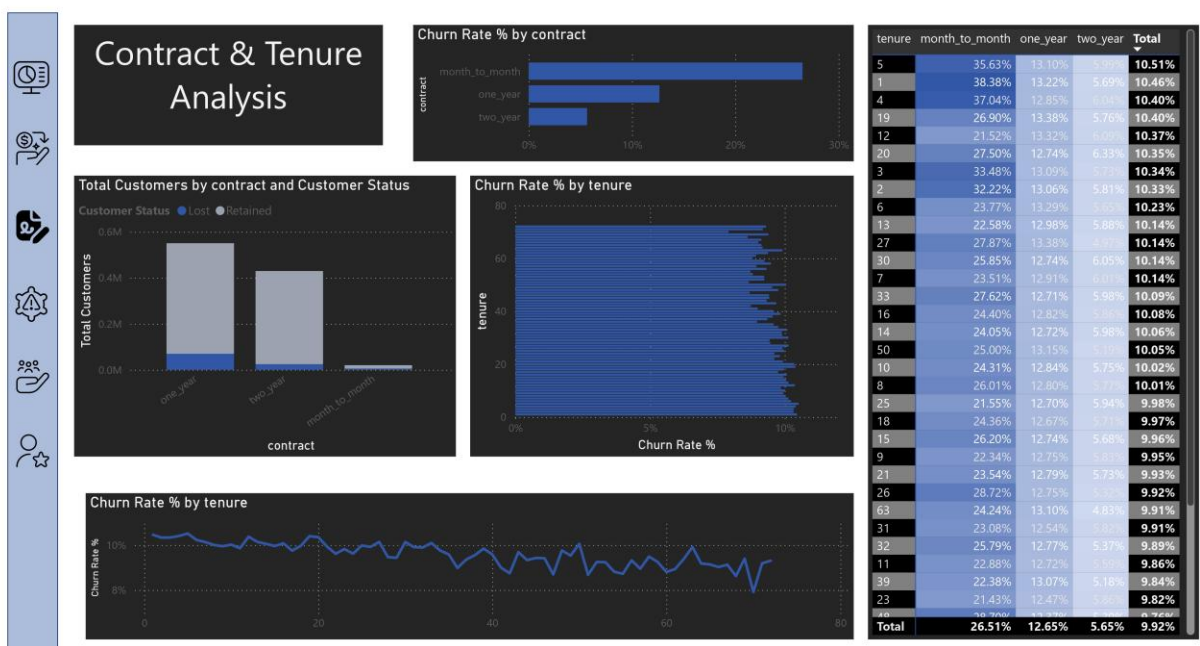


Executive Summary — 1M total customers, 99K churned (9.92%), \$8.46M lost monthly revenue, \$173.41M lifetime revenue lost, 6.16 average satisfaction.

- The churn-rate gauge highlights 9.92% against a 25% caution threshold, sitting comfortably in the green zone overall.
- The donut chart splits customers into 901K retained (90.08%) vs 99K churned (9.92%).
- The decomposition tree breaks total contract count down by number of services, letting executives drill from the 999,999 total down to any sub-segment interactively.

7. Power BI Dashboard — Contract & Tenure Analysis

This page isolates the two strongest churn predictors identified in modeling — contract type and tenure — with a churn-rate breakdown by contract, a stacked customer count by contract and status, a churn-rate-by-tenure scatter, and a full tenure lookup table.

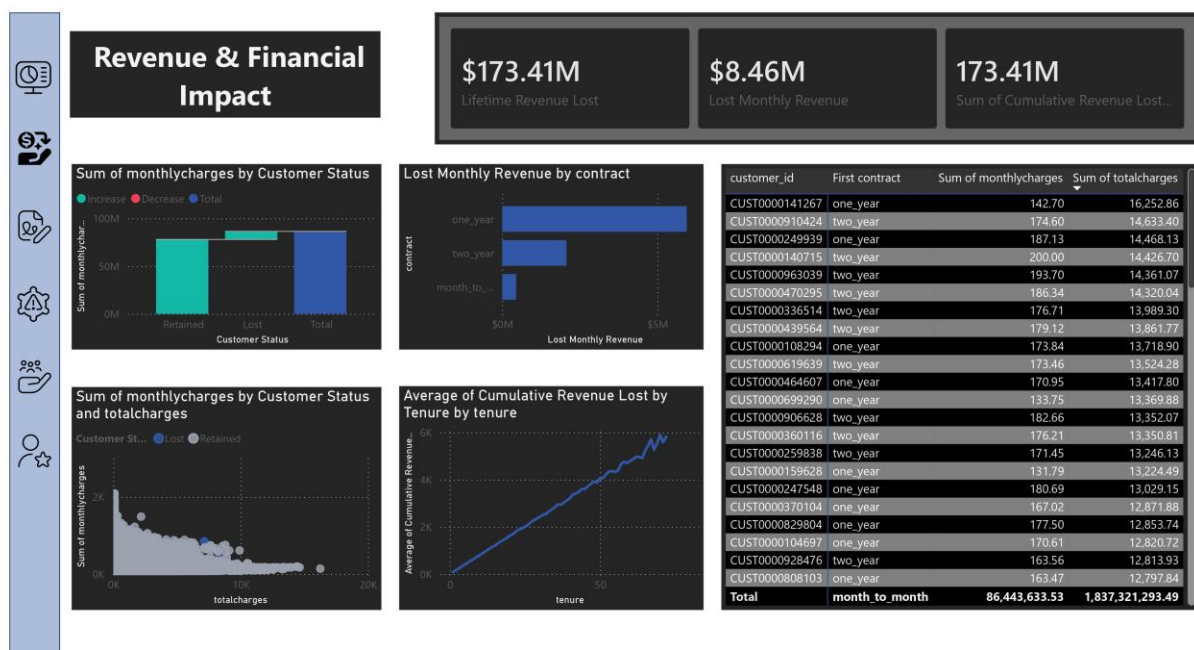


Contract & Tenure Analysis — Month-to-Month churns at roughly 26.5%, One-Year at 12.65%, Two-Year at 5.65%.

- Churn Rate % by contract confirms Month-to-Month as by far the riskiest segment, more than double One-Year and over four times Two-Year.
- Total Customers by contract and status shows One-Year and Two-Year contracts hold the bulk of the customer base with the lowest churn share.
- Churn Rate % by tenure shows an early spike in the first few months that gradually decays as tenure grows, matching the 'new customers are highest risk' insight from the EDA phase.

8. Power BI Dashboard — Satisfaction & Complaints Deep Dive

This page cross-tabulates contract type against tenure buckets, then focuses on the two behavioral signals with the clearest relationship to churn: customer satisfaction and number of complaints.

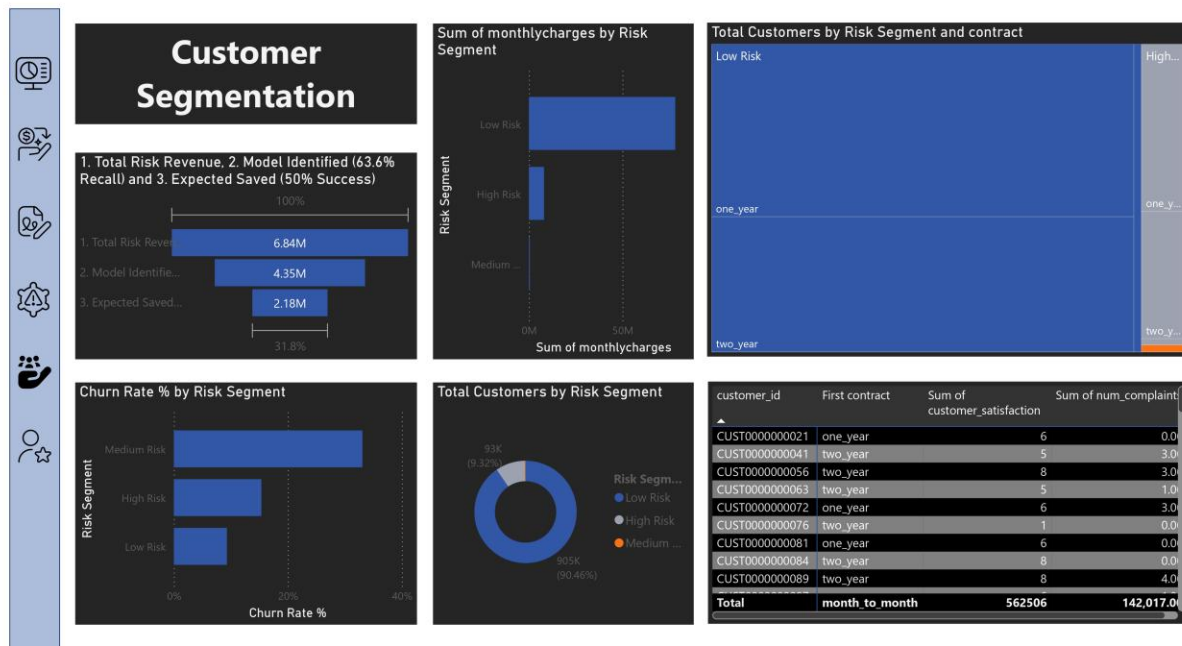


Satisfaction & Complaints Deep Dive — churn rate by contract/tenure matrix (top), and churn rate falling from ~17% to ~7% as satisfaction rises from 1 to 9 (bottom left).

- The contract × tenure matrix shows churn concentrated in the earliest tenure buckets of Month-to-Month contracts (up to 38% at 1 month tenure).
- Churn Rate % by customer_satisfaction is a clean downward staircase — the lower the satisfaction score, the higher the churn rate.
- Churn Rate % by num_complaints and contract shows complaint volume compounding with contract risk, especially for Month-to-Month customers.

9. Power BI Dashboard — Customer Behavior & Risk Signals

This page consolidates the remaining behavioral risk signals used by the model: complaint volume, late payments, tech support usage, days since last interaction, and number of subscribed services.



Customer Behavior & Risk Signals — 380K high-risk customers currently retained; churn rate climbs with late payments and falls with tech support and more services.

- A callout flags 380K customers as 'High Risk Retained' — currently retained but carrying elevated risk indicators, a priority segment for proactive retention.
- Churn Rate % by Late_Payments_Group rises steadily from 0 to 4+ late payments.
- Churn Rate % by has_tech_support shows customers without tech support churn at roughly 11.3% vs 8.6% for those with it.
- Churn Rate % by num_services shows churn generally decreasing as the number of subscribed services increases, reinforcing the bundle-upsell recommendation from the Retention Plan page.

10. Power BI Dashboard — Revenue & Financial Impact

This page translates churn into financial terms for stakeholders who think in revenue rather than churn rate, including lost monthly and lifetime revenue, a contract-level revenue breakdown, and a ranked customer-level table.



Revenue & Financial Impact — \$173.41M lifetime revenue lost, \$8.46M lost monthly revenue, with One-Year contracts contributing the largest share of lost monthly revenue.

- Sum of monthly charges by Customer Status (waterfall) shows the revenue bridge from retained to lost to total monthly charges.
- Lost Monthly Revenue by contract is highest for One-Year contracts, ahead of Two-Year and Month-to-Month — reflecting both churn rate and the larger base of One-Year customers.
- Average Cumulative Revenue Lost by Tenure rises steadily with tenure, since longer-tenured churned customers represent more lifetime value lost.
- A ranked customer table surfaces the individual highest-value churned accounts (by total charges), useful for targeted win-back outreach.

11. Power BI Dashboard — Customer Segmentation

The final Power BI page segments the customer base into Low, Medium, and High Risk groups and quantifies the model's business value: total revenue at risk, how much of it the model actually identifies (recall), and how much can realistically be saved through retention outreach.



Customer Segmentation — \$6.84M total risk revenue, \$4.35M identified by the model (63.6% recall), \$2.18M expected saved (50% success rate).

- The funnel — Total Risk Revenue → Model Identified → Expected Saved — frames the model's real-world ROI in a single visual.
- Churn Rate % by Risk Segment confirms the segmentation is well-calibrated: Medium Risk churns highest, followed by High Risk, with Low Risk churning least.
- Total Customers by Risk Segment shows 90.46% of the base sits in Low Risk, with the remaining ~9.5% split between High and Medium Risk — the actionable target population for retention campaigns.

12. Technical Implementation Notes

A few practical issues came up while building and running the dashboards, documented here for reference when handing off the app.

- **File paths:** Streamlit's working directory depends on where streamlit run is launched from, not the script's location. Paths to the dataset and model were resolved relative to the project root using `os.path.dirname(os.path.abspath(__file__))` to avoid `FileNotFoundError` when running from different terminals.
- **Scaled features in the UI:** Since the saved model was trained on scaled data, both the Prediction and Retention Plan pages use raw, human-readable slider ranges (e.g. Age 18–90, Monthly Charges \$20–120, Credit Score 300–850) and build the input row directly into the model's expected feature format before calling `predict_proba`.
- **Caching:** `@st.cache_data` and `@st.cache_resource` were used so the 1M-row dataset and the model are loaded only once per session instead of on every interaction.
- **Styling:** the dark theme, metric cards, and chart color palette across all four Streamlit pages are implemented with injected CSS rather than third-party Streamlit themes, keeping the app self-contained in a single `streamlit_app.py` file.

- **Power BI:** the six report pages share a common left-hand icon rail for navigation and a consistent dark theme, with DAX measures for churn rate, lost revenue, and risk segment reused across pages via a shared measures table.

13. Deliverables Summary

Deliverable	Description
streamlit_app.py	Full dashboard application — Overview, EDA, Prediction, and Retention Plan pages
Overview Page	Headline churn metrics and key insight cards
EDA Page	Interactive charts reproducing the core EDA findings
Prediction Page	Live churn probability scoring with risk factor explanation
Retention Plan Page	Risk factor breakdown plus prioritized, actionable retention recommendations
Power BI Report (.pbix)	6-page workbook: Executive Summary, Contract & Tenure, Satisfaction & Complaints, Customer Behavior & Risk, Revenue & Financial Impact, Customer Segmentation

End of Dashboards Report